```
import cv2
import numpy as np
from google.colab import files
import matplotlib.pyplot as plt

# Upload image
uploaded = files.upload()

# Get uploaded file name
image_path = list(uploaded.keys())[0]

# Read image
img = cv2.imread(image_path)

# Check if image is loaded
if img is None:
    print("Error: Image not found!")
else:
    # Get image dimensions
    rows, cols = img.shape[:2]

    # Source points
    src_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, rows - 1],
        [cols - 1, rows - 1]
    ])

    # Destination points
    dst_points = np.float32([
        [0, 0],
        [cols - 1, 0],
        [0, int(0.7 * rows)],
        [cols - 1, int(0.7 * rows)]
    ])

    # Compute homography matrix
    M, _ = cv2.findHomography(src_points, dst_points)

    # Apply homography transformation
    homography_img = cv2.warpPerspective(img, M, (cols, rows))

    # Save output image
    cv2.imwrite("Transformation_Using_Homography_Image.jpg", homography_img)

    # Convert BGR to RGB for display
    original_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    homography_rgb = cv2.cvtColor(homography_img, cv2.COLOR_BGR2RGB)
```

```
# Display images
plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.imshow(original_rgb)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(homography_rgb)
plt.title("Homography Transformation")
plt.axis("off")

plt.show()

# Download the transformed image
files.download("Transformation_Using_Homography_Image.jpg")
```

Choose Files sampled_image.jpg

sampled_image.jpg(image/jpeg) - 9699 bytes, last modified: 7/22/2026 - 100% done

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Saving sampled_image.jpg to sampled_image (5).jpg
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